AI Agents Unleashed

Patterns, Prototypes & Production

Workshop by

Tezan Sahu

services.tezansahu@gmail.com

A little about me...

- Microsoft IDC (2021 – Present)
  - Software Engineer II (2024 – Present) *[Enterprise Copilot Extensibility]*
  - Data & Applied Scientist II (2023 - 2024) *[Bing Autosuggest]*
  - Data & Applied Scientist (2021 - 2023) *[Bing Autosuggest]*
- IIT Bombay (2017 – 21)
  - Major in Mechanical Engineering *(9.90 CPI)*
  - Minor in Computer Science & Engineering
  - Institute Rank 2 | Department Rank 1
- Author of "Beyond Code" – a National Bestseller
- Holder of 2 US Patents
- Top rated AI & Data Science Mentor
- Empowered 20,000+ students & professionals in AI & Data Science
  - Through tech talks, webinars & workshops across several top colleges & ed-tech platforms

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Agenda

- AI Agent Fundamentals
- Agent Architecture & Design Patterns
- Rise of Agentic Protocols
- Introduction to Microsoft’s AutoGen
- [Live Demos] Building Single-Agent Systems
- [Live Demos] Building Multi-Agent Systems
- Taking AI Agents to Production

AI Agent Fundamentals

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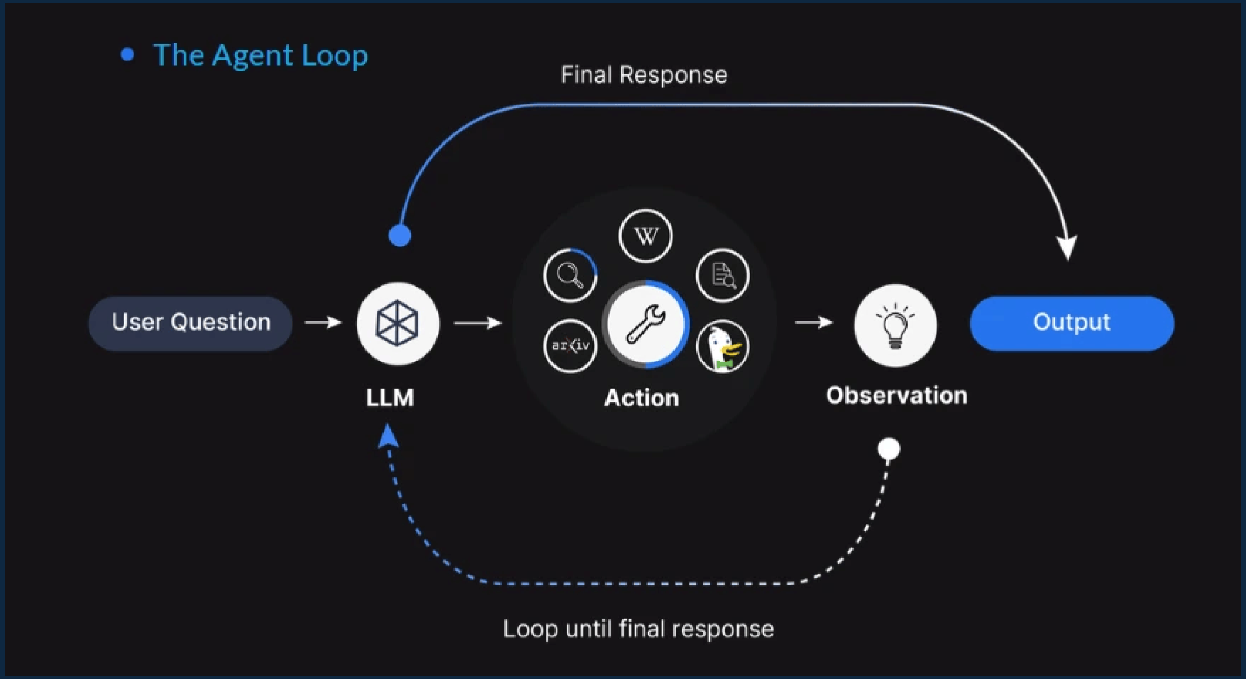
What's an AI Agent?



An AI agent is a system or program
capable of autonomously performing tasks
on behalf of a user or another system
by designing its workflow and utilizing available tools

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What Makes Agents So Smart?

- Autonomy: Operates without constant human oversight
- Goal-Orientation: Works towards achieving defined objectives
- Context-Awareness: Adapts to changes in the environment or requirements
- Interactivity: Communicates with humans and other systems seamlessly

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LLMs – The Brains Behind Agents

Large Language Models (LLMs) are AI models trained on massive datasets to understand and generate human-like text

Core Capabilities

✅ Language understanding

✅ Text generation

✅ Summarization, translation, and Q&A

✅ Few-shot and zero-shot learning

How do they work?

▶️ Token-based processing using transformer architecture

▶️ Predict next token based on context

▶️ Fine-tuning and prompt engineering enhance specificity

Limitations

🚫 Lacks real-time knowledge (cutoff dates)

🚫 Prone to hallucination (confident but incorrect)

🚫 No direct interaction with external systems

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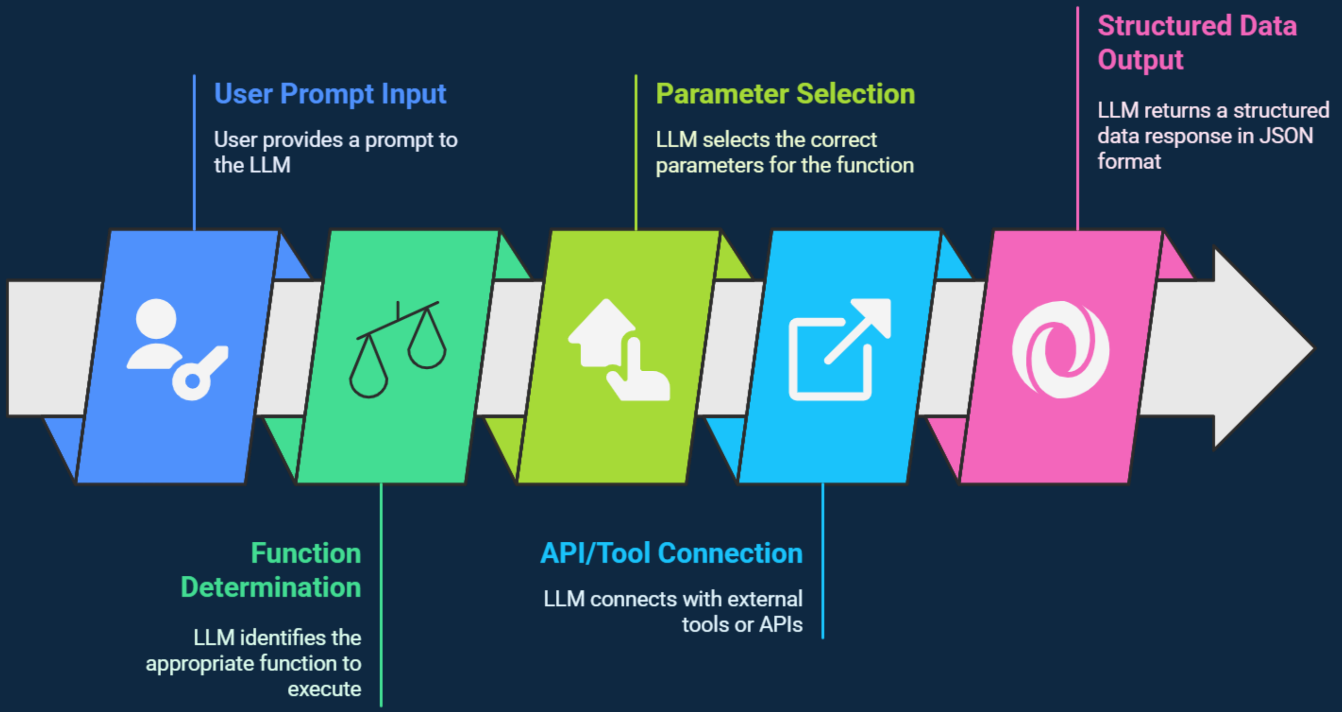
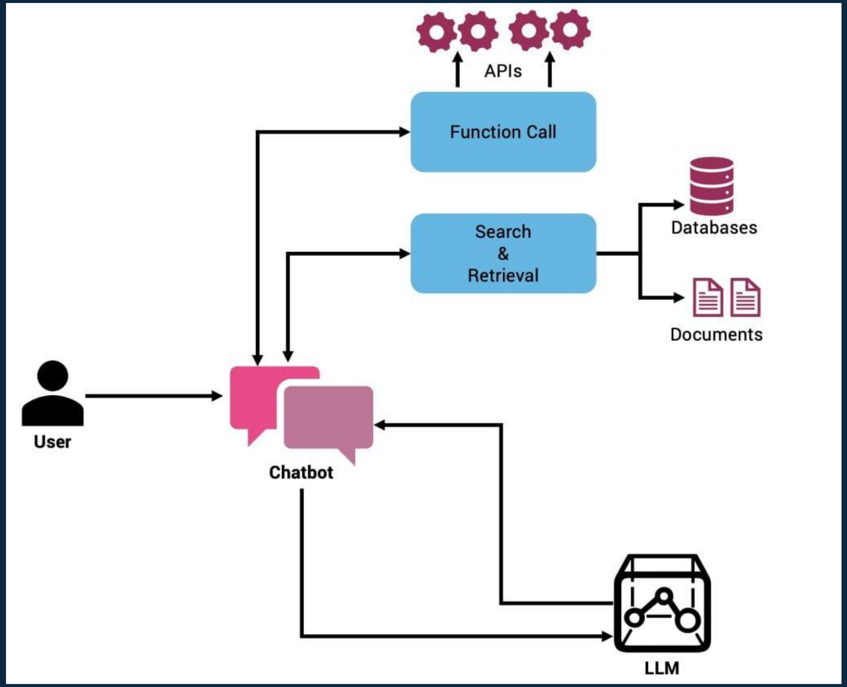
Function Calling in LLMs

LLM's ability to determine, from a user prompt, the appropriate function to execute from available options and the correct parameters to pass to that function





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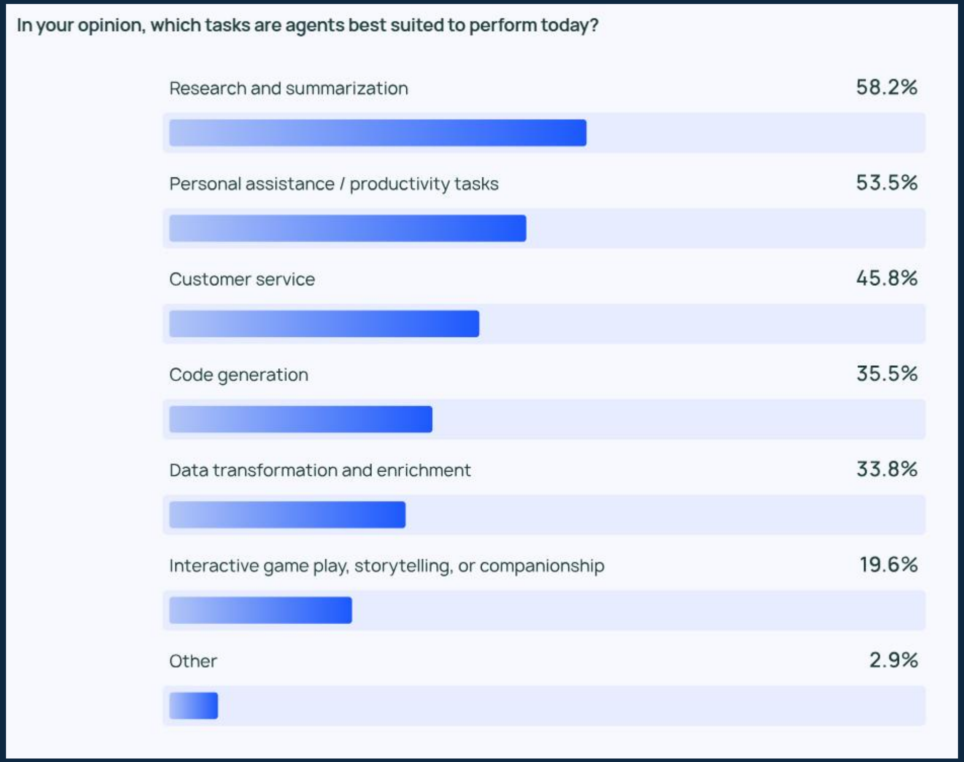
What Can AI Agents Do Today?

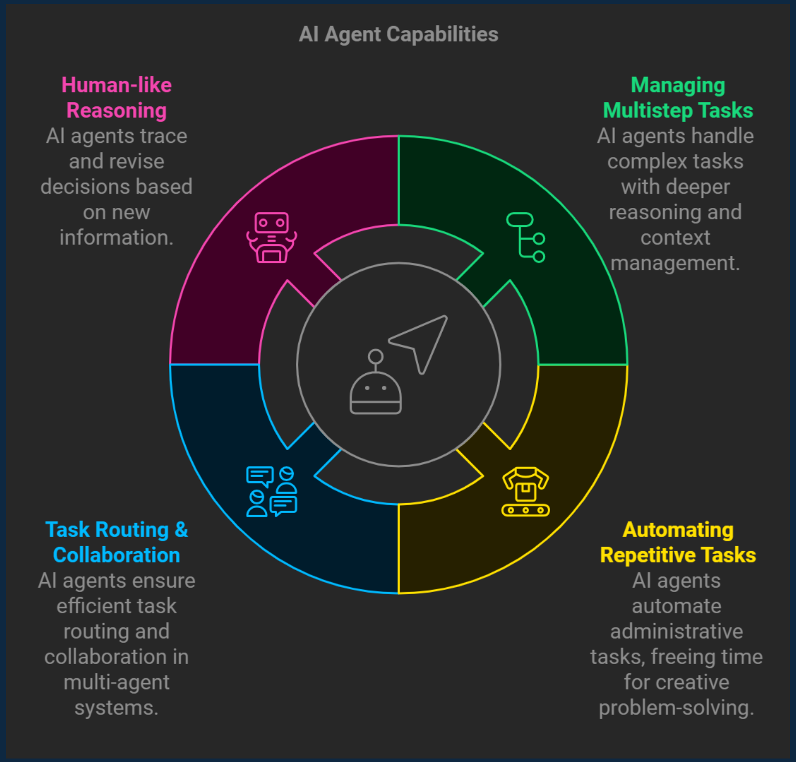
51.1%

Companies have Agents in production

90%

Respondents from Non-Tech Companies have or plan to have Agents in production





[Ref] State of AI Agents Report - LangChain

Automation vs Workflows vs Agents

	Automation	AI workflow	AI agent
Definition	A program that executes predefined, rule-based tasks automatically	A program that calls an LLM via API for one or more steps	A program designed to perform non-deterministic tasks autonomously
Core foundations	Boolean logic	Boolean logic, Fuzzy logic	Fuzzy logic, Autonomy
Tasks	Deterministic, predefined tasks	Deterministic tasks requiring flexibility	Non-deterministic, adaptive tasks
Strengths	• Delivers reliable outcomes • Fast to execute	• Better handling of complex rules • Great for pattern recognition	• Highly adaptive to new variables • Simulates human-like behavior and reasoning
Weaknesses	• Limited to tasks explicitly programmed • Cannot adapt to new scenarios • Struggles with complexity	• Requires data to train models effectively • Harder to debug and interpret	• Less reliable, may produce unpredictable undesired outcomes • Slower to execute
Example	Send a Slack notification every time a new lead signs up on our website	Analyze, score and route every website inbound lead using ChatGPT	Perform a full internet search on every inbound lead and update infos

Source: Tyler McGregory via LinkedIn

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Automation vs Workflows vs Agents

Which solution fits my business?

AI Workflows

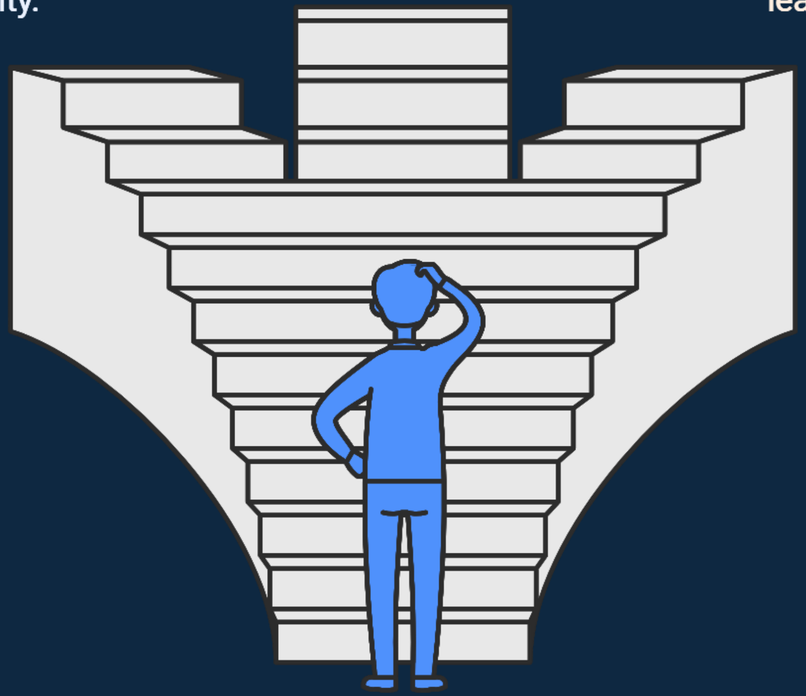
Ideal for complex processes requiring dynamic decision-making and scalability.

Automations

Suitable for repetitive tasks with predefined rules and limited flexibility.

AI Agents

Best for autonomous tasks that require real-time adaptation and learning.



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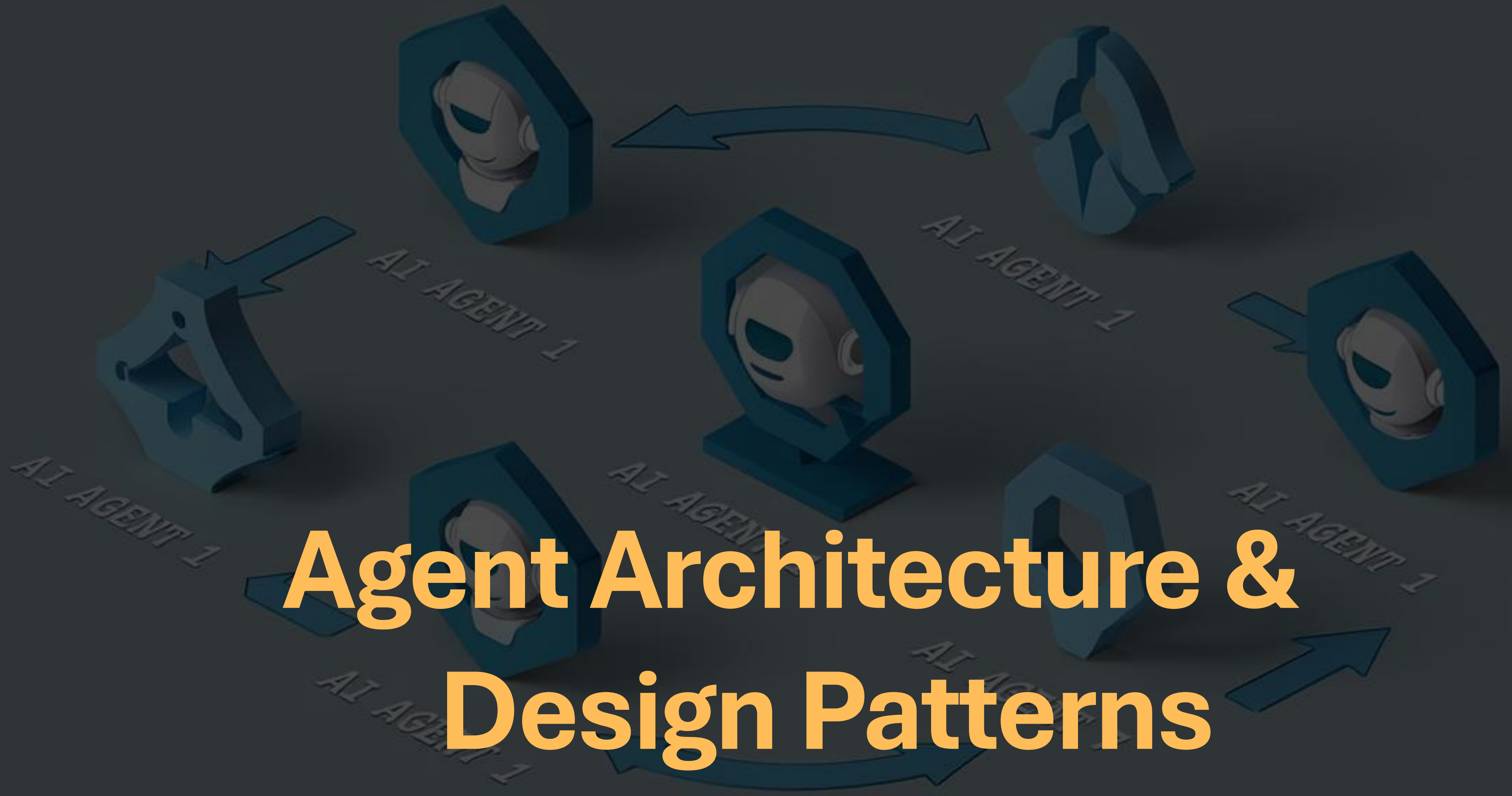

Agent Architecture & Design Patterns

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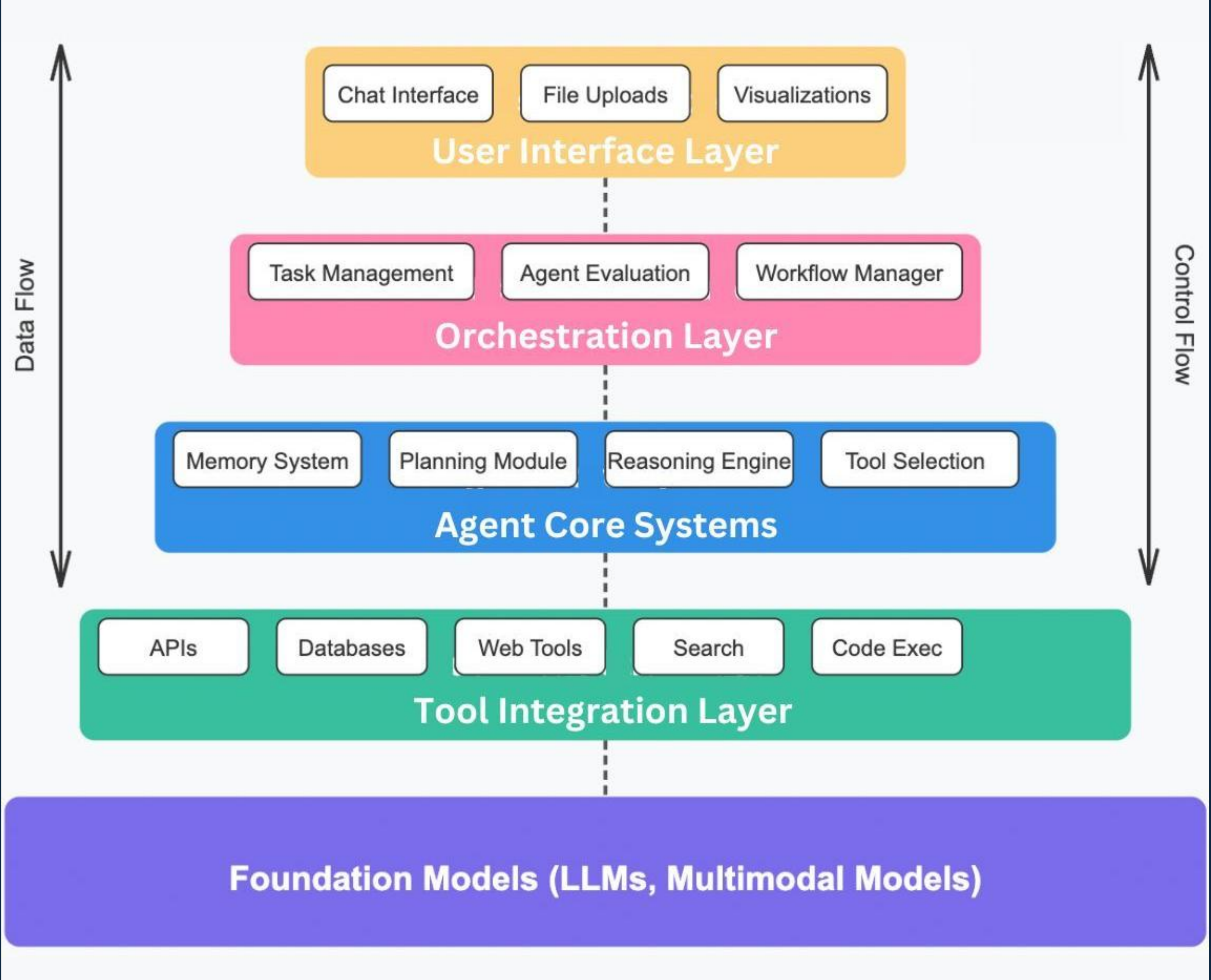
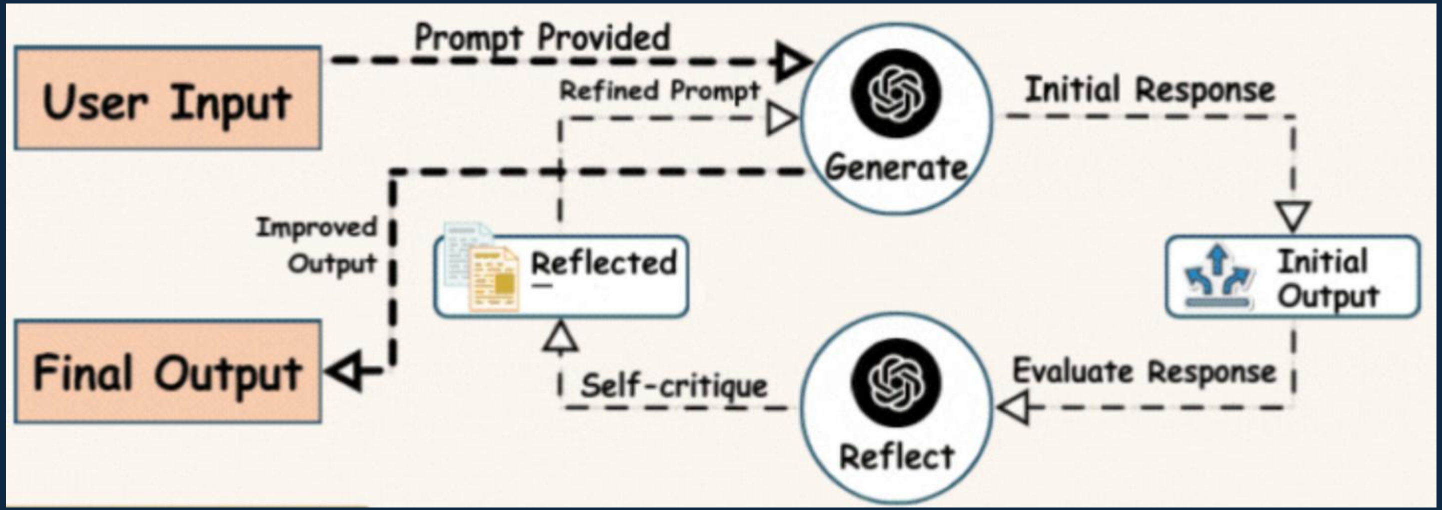
Technical Architecture of AI Agents

A 30,000-ft view…



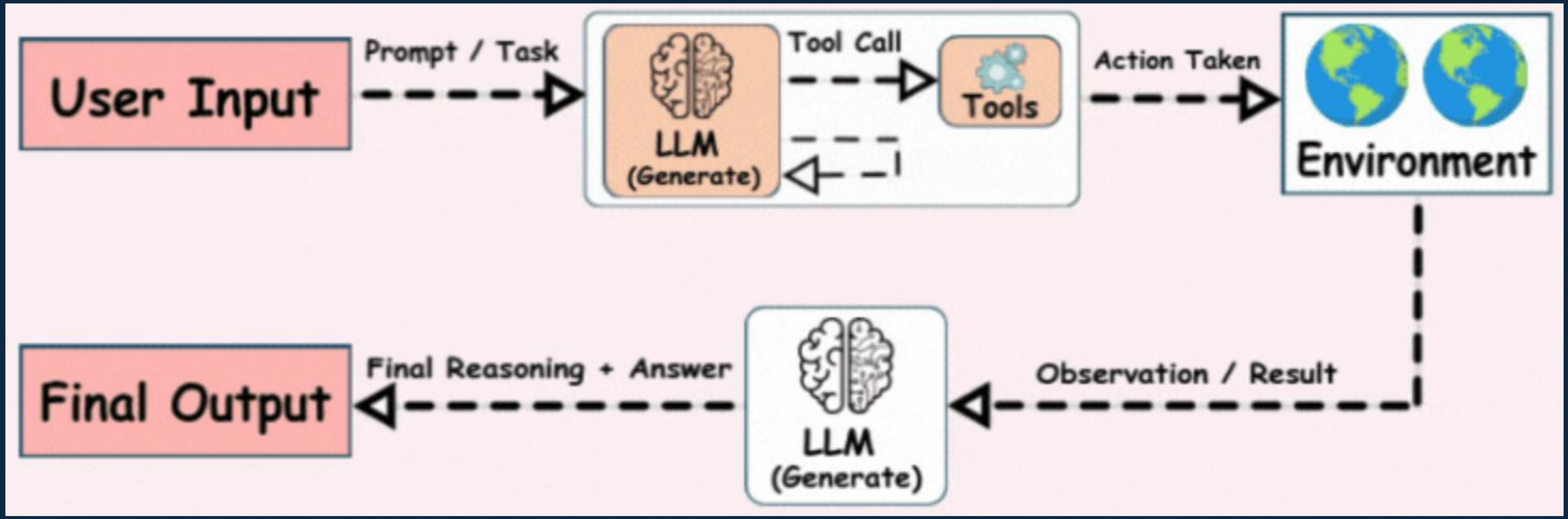
Reflection Pattern – Teach Your Agent to Self-Reflect

- **Key Features:**
  - Self-Evaluation: Agents assess their performance using metrics or goals
  - Feedback Loops: They adjust their strategies based on the evaluation
- **Scenario:**
  - A recommendation agent reflects on user feedback to improve its suggestions



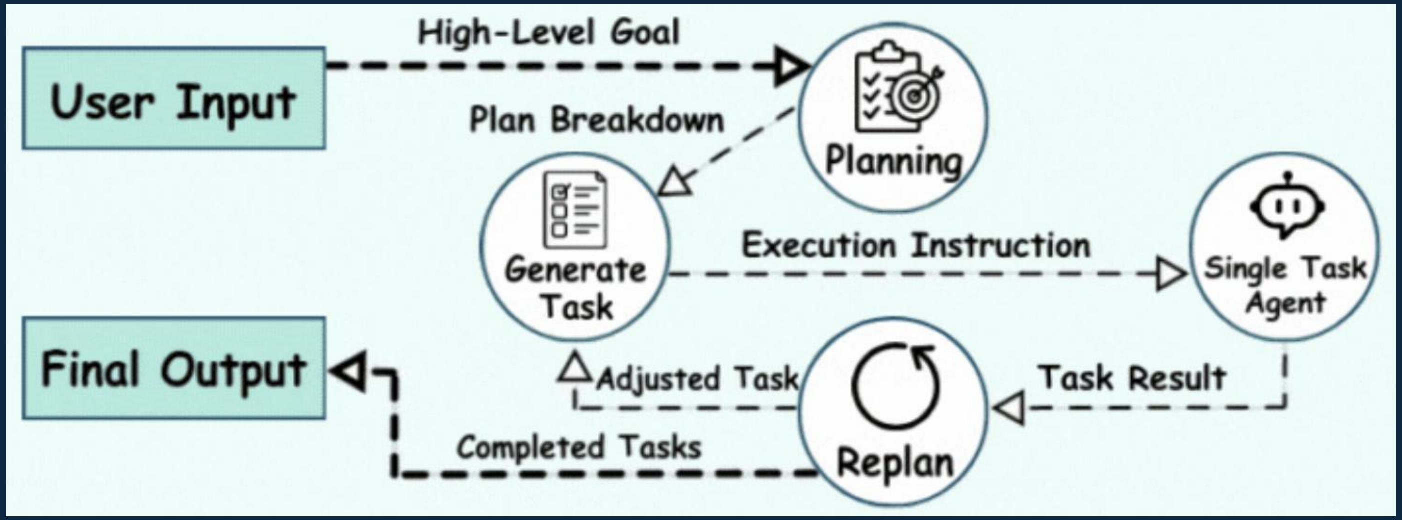
ReAct Pattern – Handing AI Its Toolkit

- **Key Features:**
  - Reason & Act: Agents reasons about what to do & then calls tool(s) accordingly
  - Adaptation: Once the tool is executed, it observes the outcome & repeats until it can generate final response
- **Scenario:**
  - When asked to compare stock prices of 2 companies, the agent reasons that it needs 2 tool calls, calls the APIs, reads the response & gives final comparison



Planning Pattern – Making AI Think Before It Acts

- **Key Features:**
  - Dynamic Goal Alignment: Plans are generated in real-time based on inputs or changes
  - Sequential Execution: Agents execute plans step-by-step, ensuring dependencies are managed
- **Scenario:**
  - A logistics agent plans delivery routes by optimizing for distance, fuel cost, and traffic conditions



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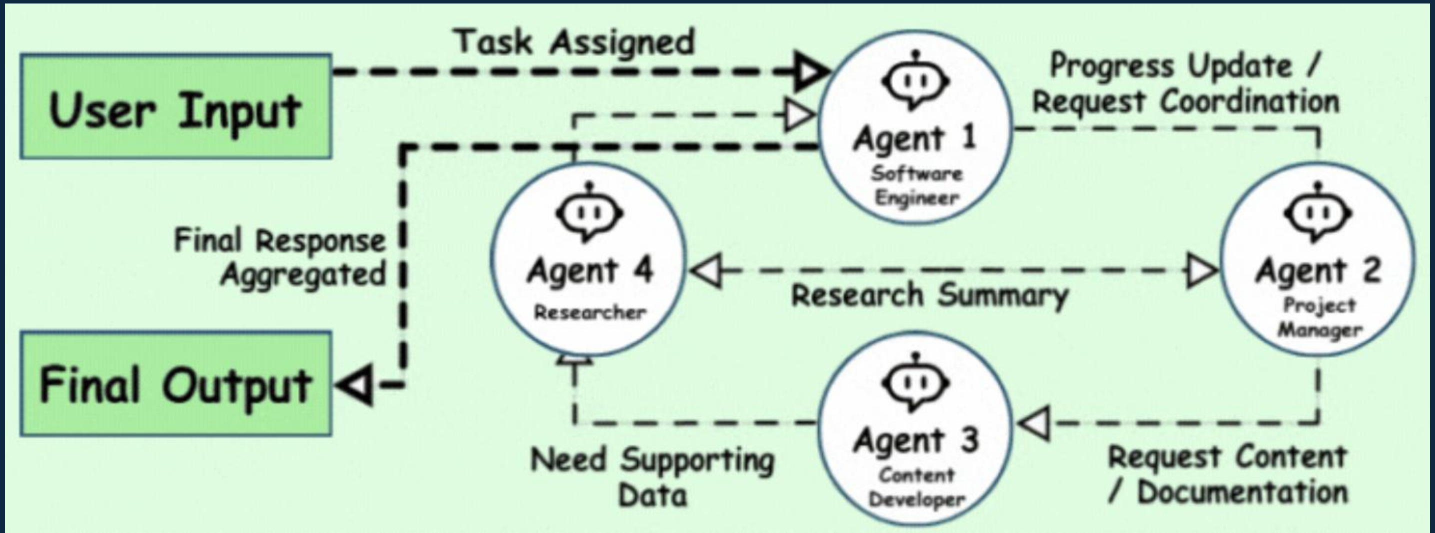
Multi-Agent Pattern – Teamwork Makes Dream Work

- **Key Features:**
  - Division of Labor: Agents handle different aspects of a task
  - Coordination: A central orchestrator ensures agents align their actions
- **Scenario:**
  - A marketing workflow involves one agent for content generation, another for scheduling, and a third for analytics



Rise of Agentic Protocols

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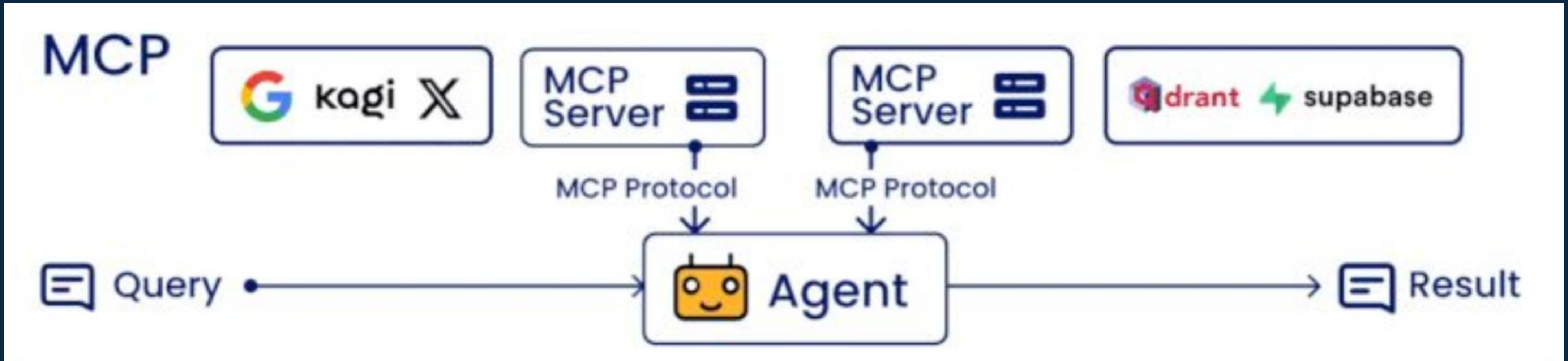
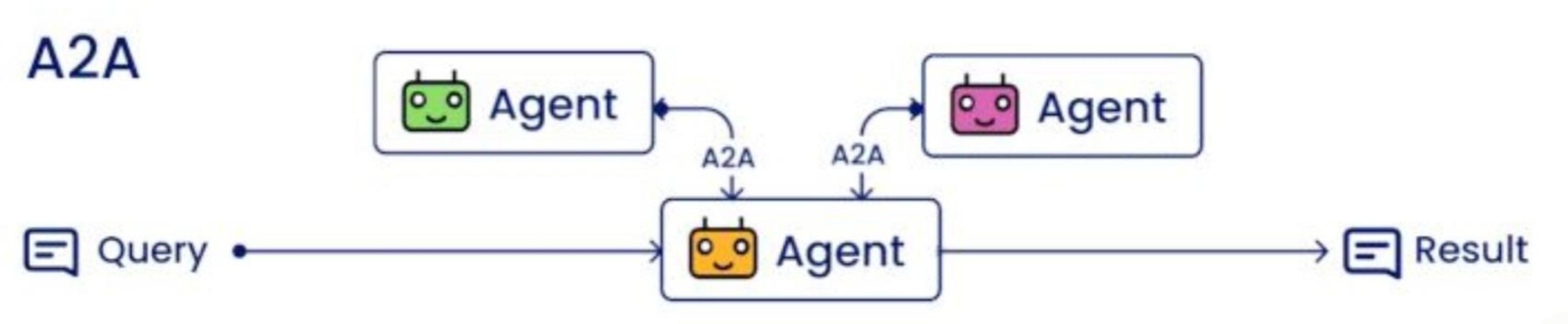
Model Context Protocol (MCP)

- Provides a structured way for agents to access tools
  - Think: *USB-C for Tools*
- Developed by Anthropic



Agent-to-Agent Protocol (A2A)

- Enables communication & collaboration between AI agents
- Developed by Google



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MCP vs A2A

FEATURE	MCP	A2A
Primary Focus	Tool use and context provision	Agent discovery and communication
Communication Pattern	Model-to-tool, contextual	Agent-to-agent, message-based
Protocol Structure	Client-server model	Discovery and messaging system
Core Components	Hosts, Servers, Clients	Agent cards, Skills, Communication methods
Primary Use Case	Enhancing model capabilities	Coordinating multiple agents
Implementation Complexity	Moderate	More complex

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Introduction to Microsoft's AutoGen

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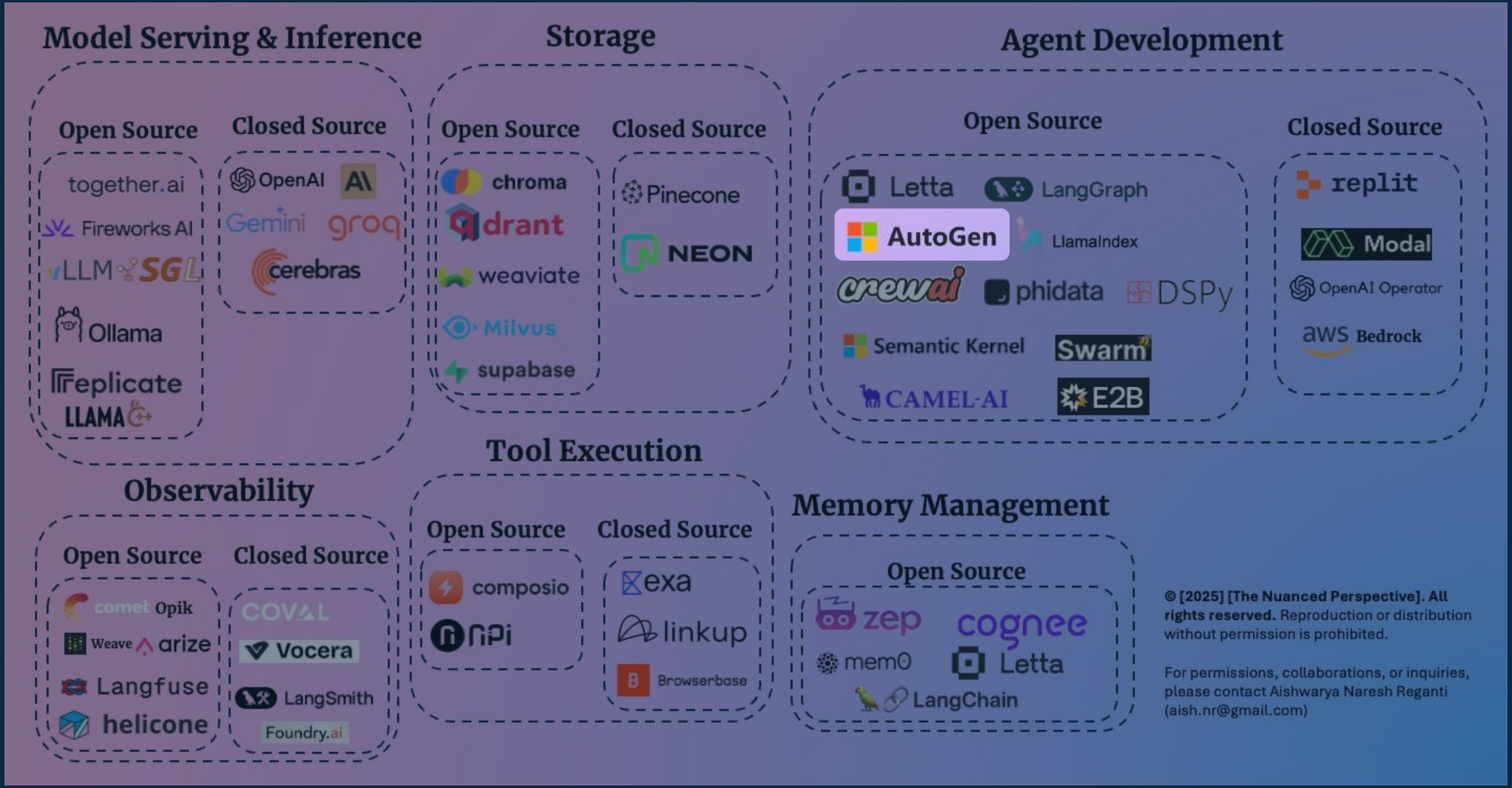
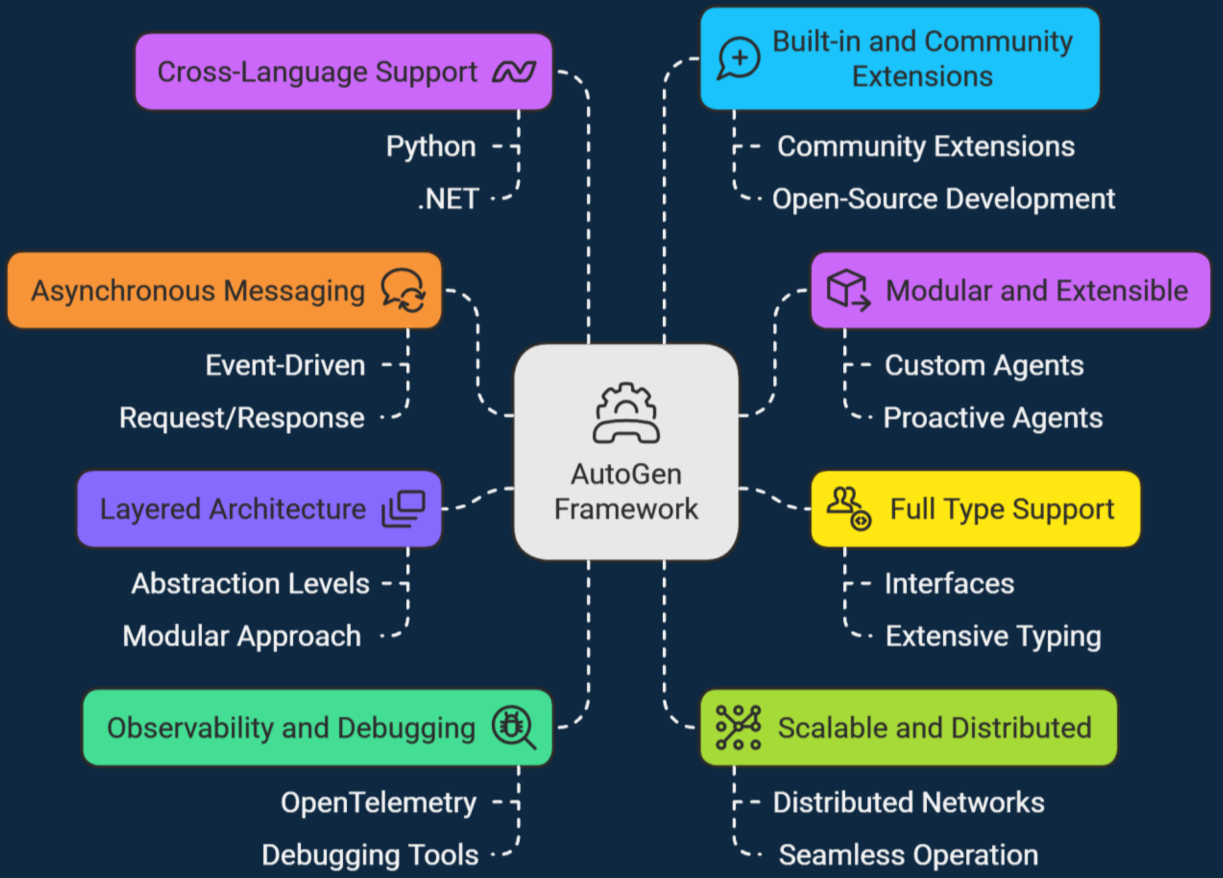
The AI Agent Stack – Jan 2025



Fundamentals of AutoGen

Open-Source framework for building AI agents & facilitating cooperation among multiple agents to solve tasks



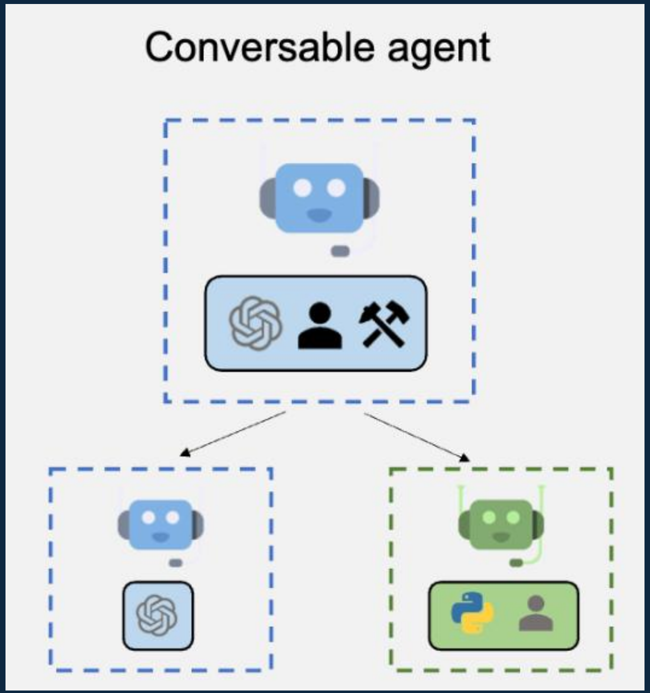
- [GitHub Repository]
- [Documentation]

AgentChat Fundamentals – Agents

AssistantAgent

This is a general-purpose agent powered by an LLM.

It can engage in natural language conversations, generate plans, and provide instructions.



UserProxyAgent

This is a special agent that acts as a proxy for the end-user.

It can relay information between the user and other agents

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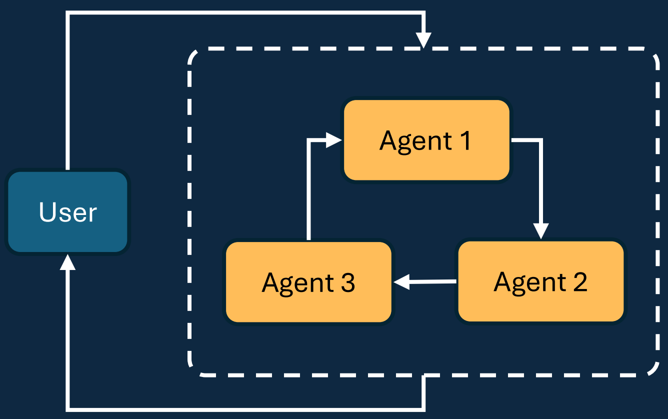
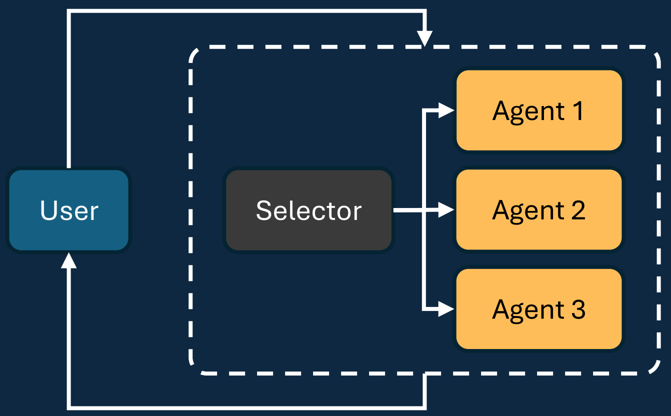
AgentChat Fundamentals – Teams

Centralized





Round Robin Group Chat

All agents share the same context and take turns responding in a round-robin fashion.

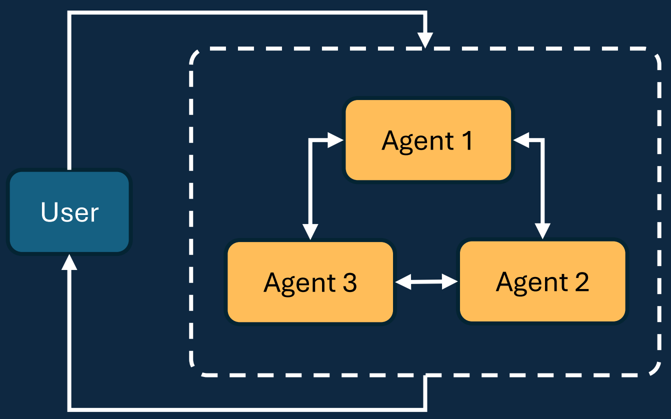
Each agent, during its turn, broadcasts its response to all other agents

Selector Group Chat

Participants take turns broadcasting messages to all other members.

An LLM selects the next speaker based on the shared context, enabling dynamic, context-aware collaboration

Decentralized



Swarm

A team in which agents can “hand off” tasks to other agents based on their capabilities.

Agents make local decisions about task planning, rather than relying on a central orchestrator

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AgentChat Fundamentals – Messages [Link]

Agent-Agent Messages

Includes message types for agent-to-agent communication

- TextMessage
- MultiModalMessage
- HandoffMessage

Internal Events

Includes messages used to communicate events and information on actions within the agent itself

- ToolCallRequestEvent
- ToolCallExecutionEvent

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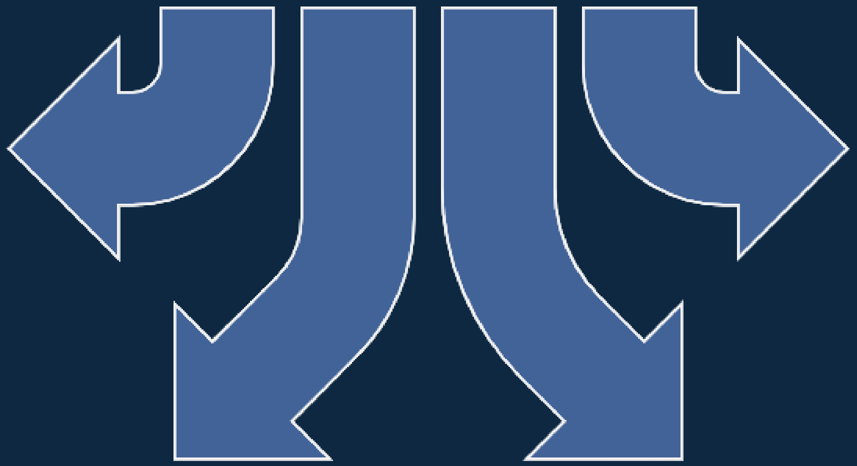
AgentChat Fundamentals – Termination [Link]

Common Termination Conditions



MaxMessageTermination

Terminates after a set number of messages, ensuring limit-based control.

TextMentionTermination

Stops when specific text is detected, allowing for keyword-based control.

HandoffTermination

Pauses for user input during a handoff, facilitating interactive control.

TextMessageTermination

Ends when a text message is sent, focusing on message-based control.

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Time for some Live Action!

Let's Get Your Environment Ready...

- Clone the repository: tezansahu/building-ai-agents-with-autogen
    - Checkout aaruush_2025 branch
- Create Virtual Environment
- Install all required packages
- Set up GitHub Personal Access Token (PAT)
    - Access LLMs (GPT-4o, GPT-4o-Mini, etc.) for FREE
- Get API Keys for Tools
    - Serper
    - Firecrawl
    - Typefully

Building Single-Agent Systems

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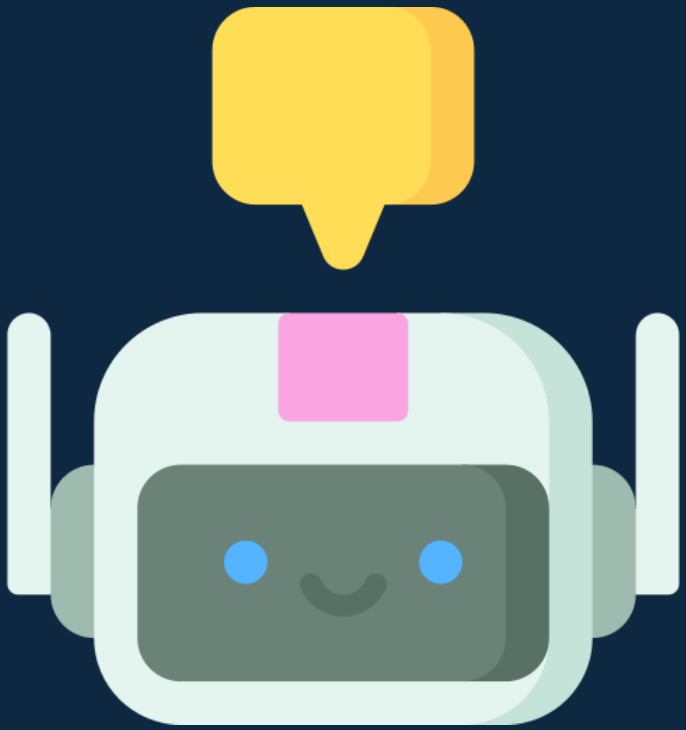
Basic Single Agent – Instructions Only



Career Mentor Agent

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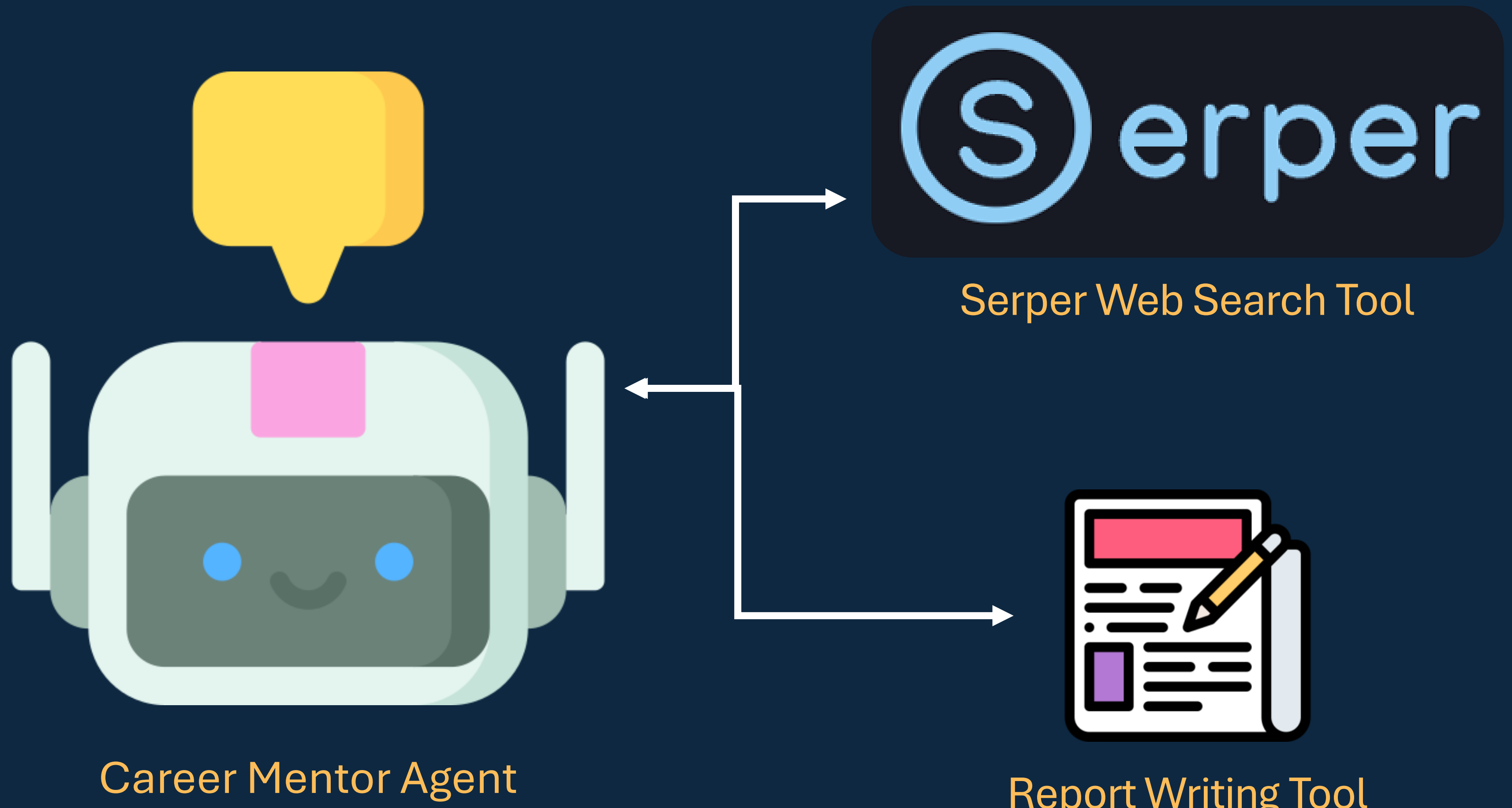
Single Agent with Tools



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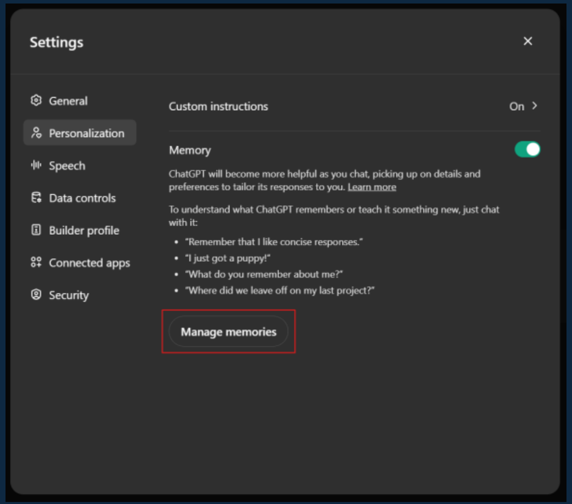
Memory in AI Agents

Why Memory?

- It may be valuable to maintain a store of useful facts that can be intelligently added to the context of the agent just before a specific step.

- It allows agents to build knowledge incrementally, reference past interactions appropriately, and maintain contextual awareness across multiple sessions.

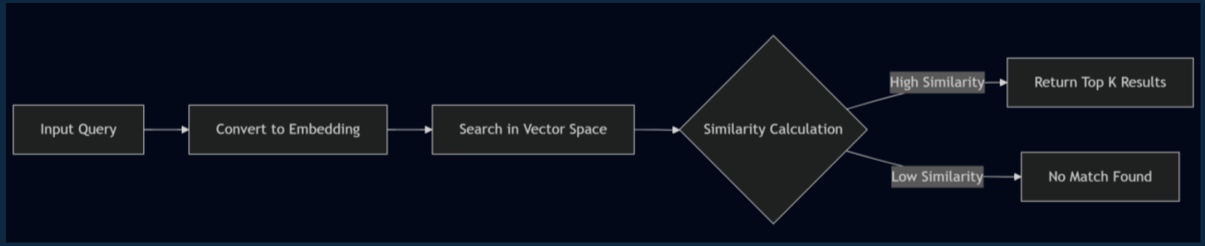


Memory in ChatGPT

Vector DB for Memory

Vector store (e.g.: ChromaDB) memory provides:

- Persistent storage beyond individual sessions
- Semantic search capabilities for finding relevant past information
- Efficient storage and retrieval of large amounts of information



Semantic Retrieval from VectorDB

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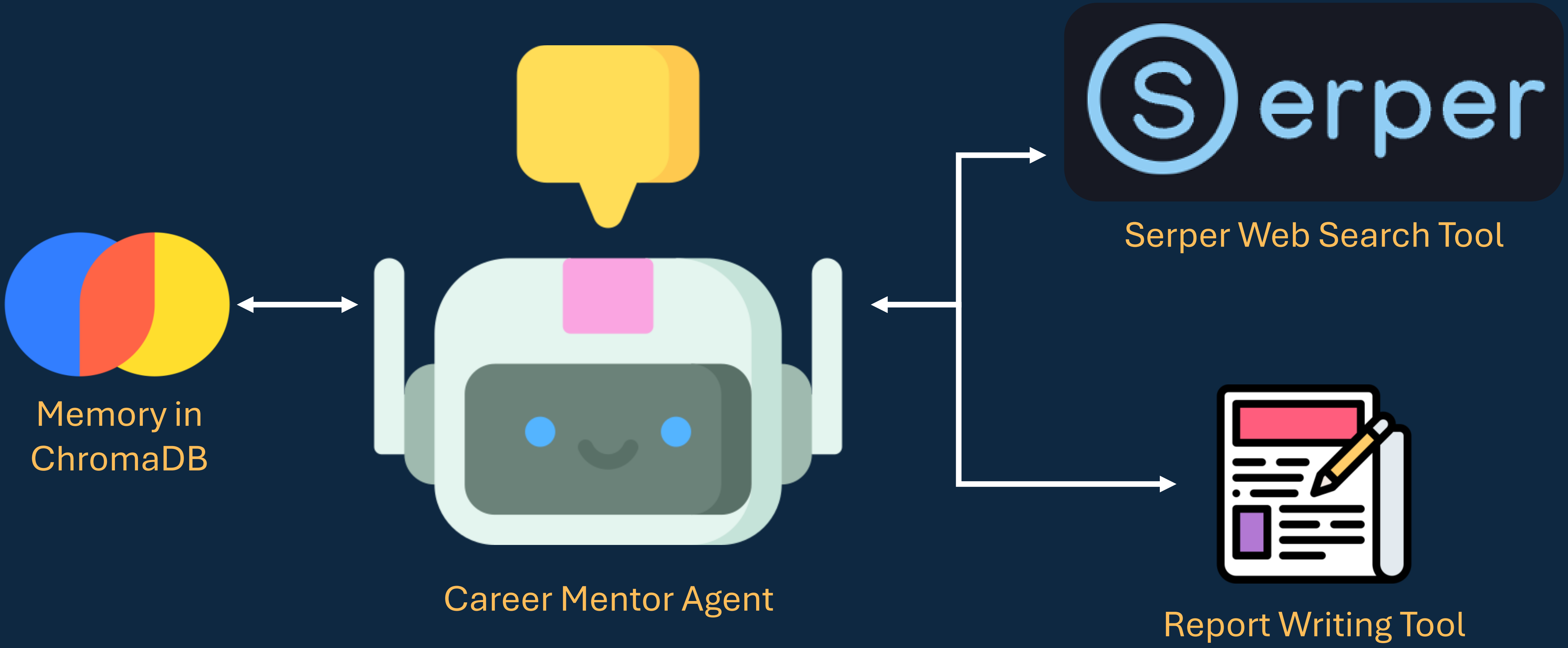
Single Agent with Tools & Memory



Agentic RAG

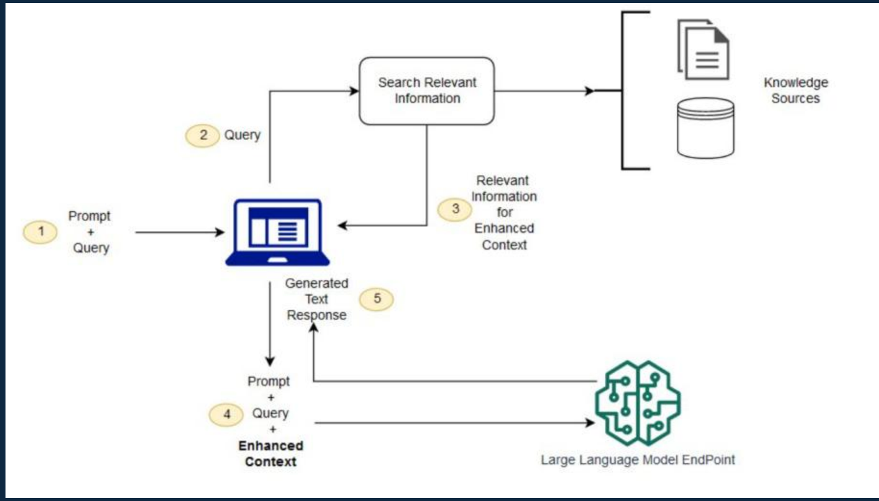
What is RAG?

- Retrieval-Augmented Generation (RAG) is a powerful approach that enhances LLM outputs by retrieving relevant information from external knowledge bases

- The RAG process involves:
  - Converting user queries into vector embeddings
  - Searching a vector database for semantically similar documents
  - Retrieving relevant documents and incorporating them into the prompt context
  - Generating responses that leverage both the model's general knowledge and the retrieved information

Why RAG using LangChain?

- AutoGen does NOT have built-in RAG Tool
- AutoGen allows tools built with LangChain to be adapted
- LangChain offers variety of:
  - Document Loaders
  - Chunking Strategies
  - Integration with ChromaDB

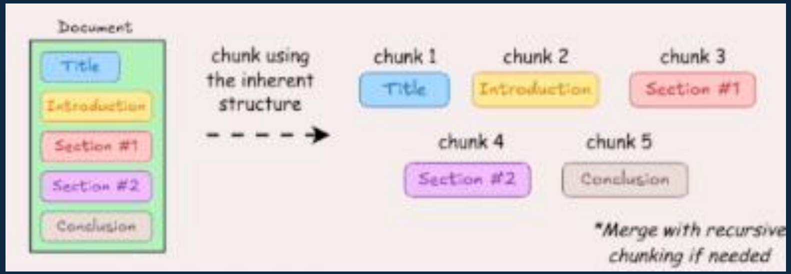


Chunking Strategies for RAG

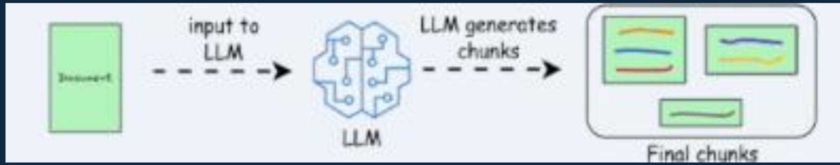
Simple Strategies

Fixed-Length Chunking

Document Structure-Based Chunking

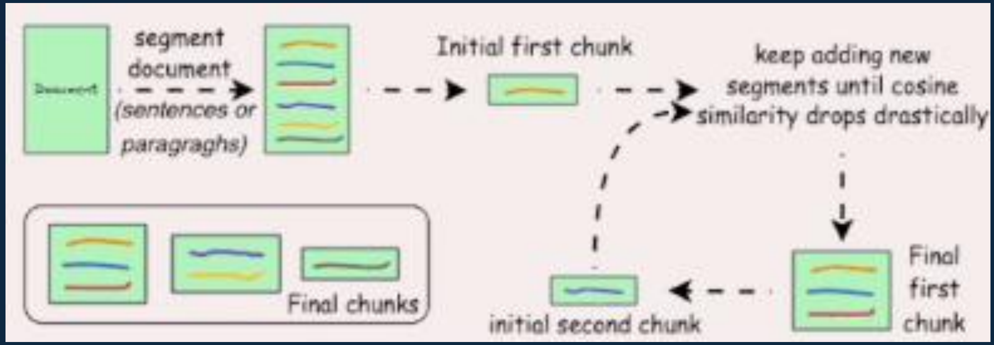


LLM-Based Chunking

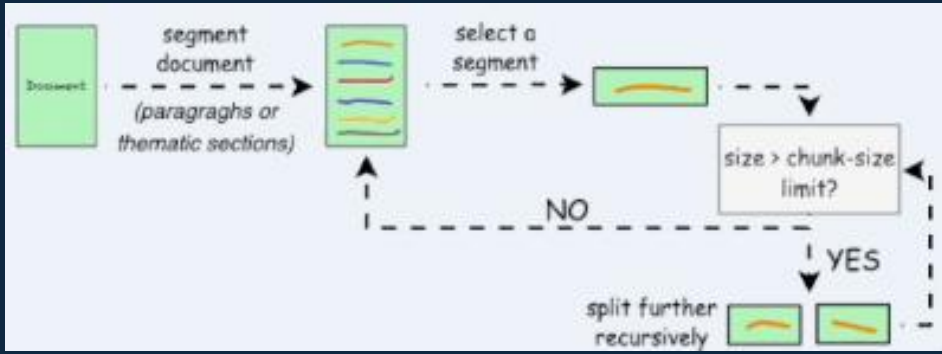


Advanced Strategies

Semantic Chunking



Recursive Chunking



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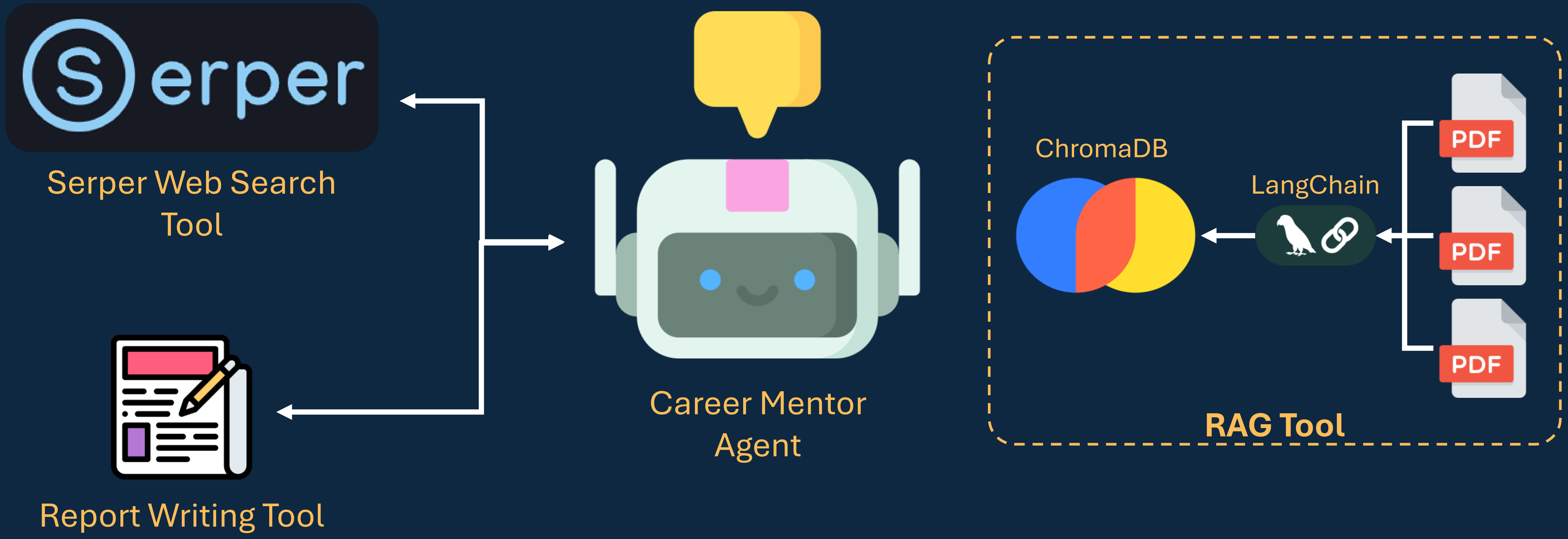
Single Agent with Tools & RAG



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Building Multi-Agent Systems

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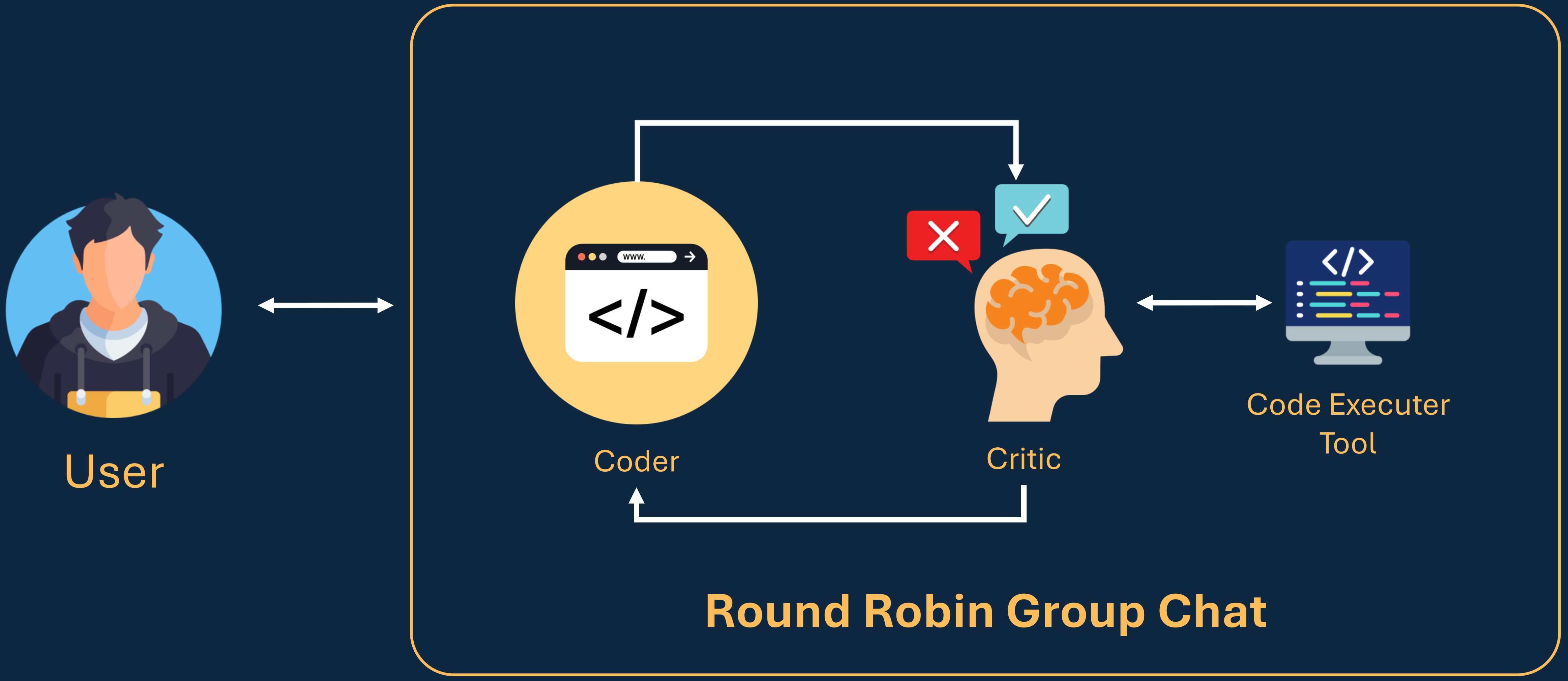
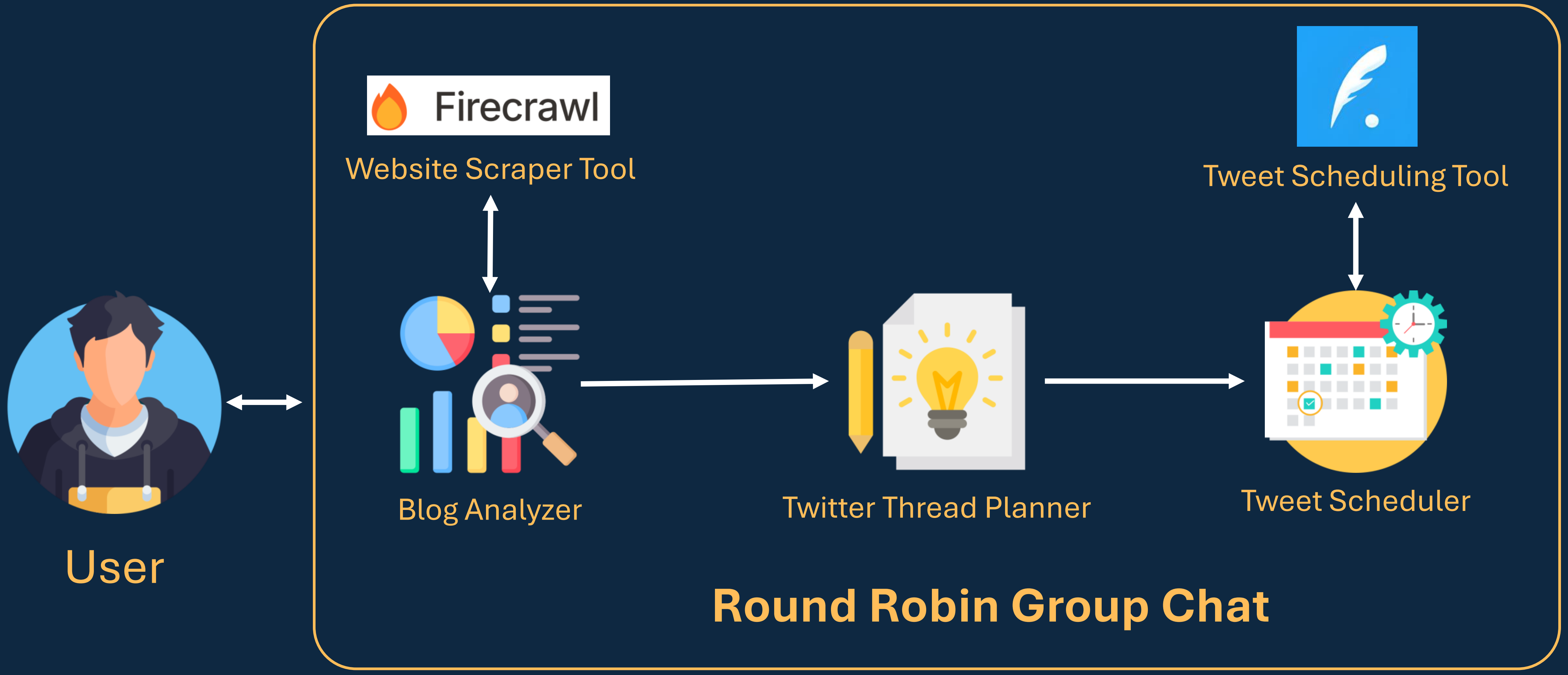
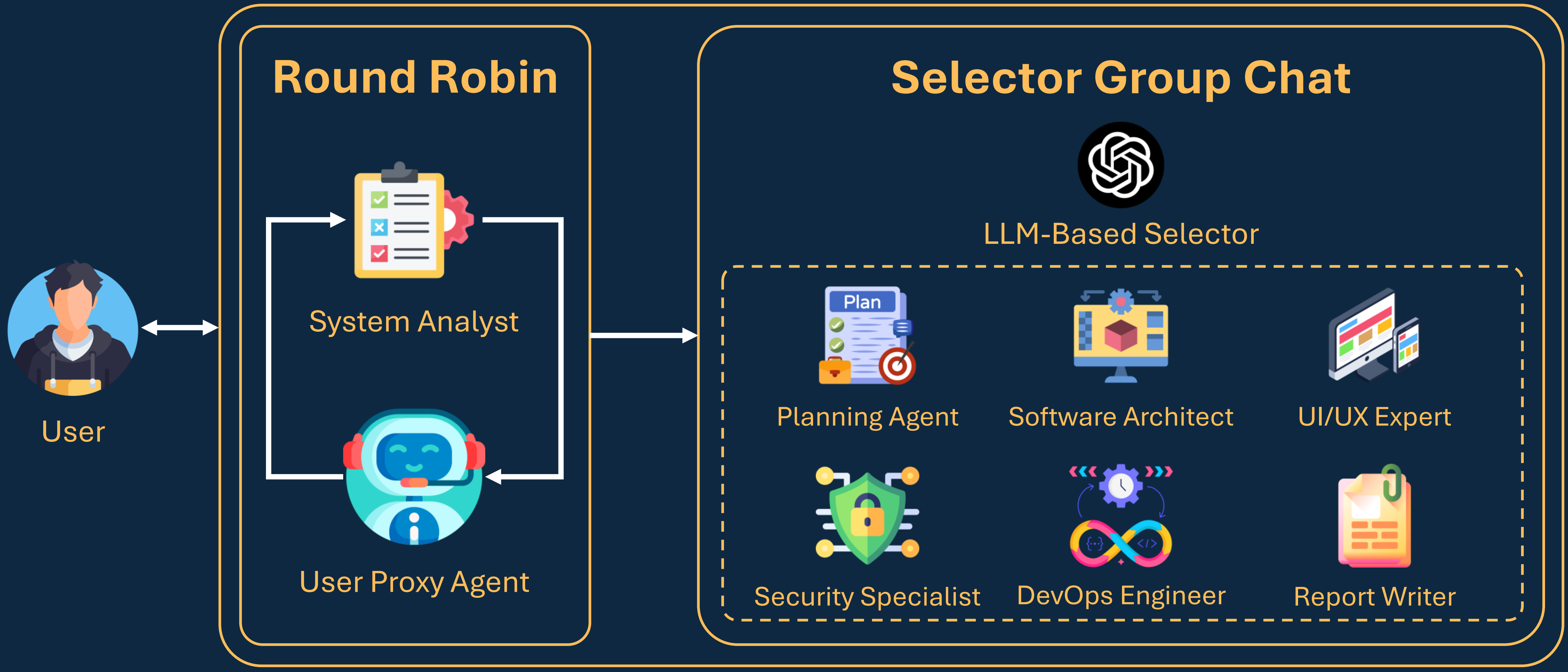
Coder-Reviewer Agents – Reflection Pattern



Blog-to-Twitter Thread Scheduler – Sequential



System Design Agent – Human-in-the-Loop + Planning



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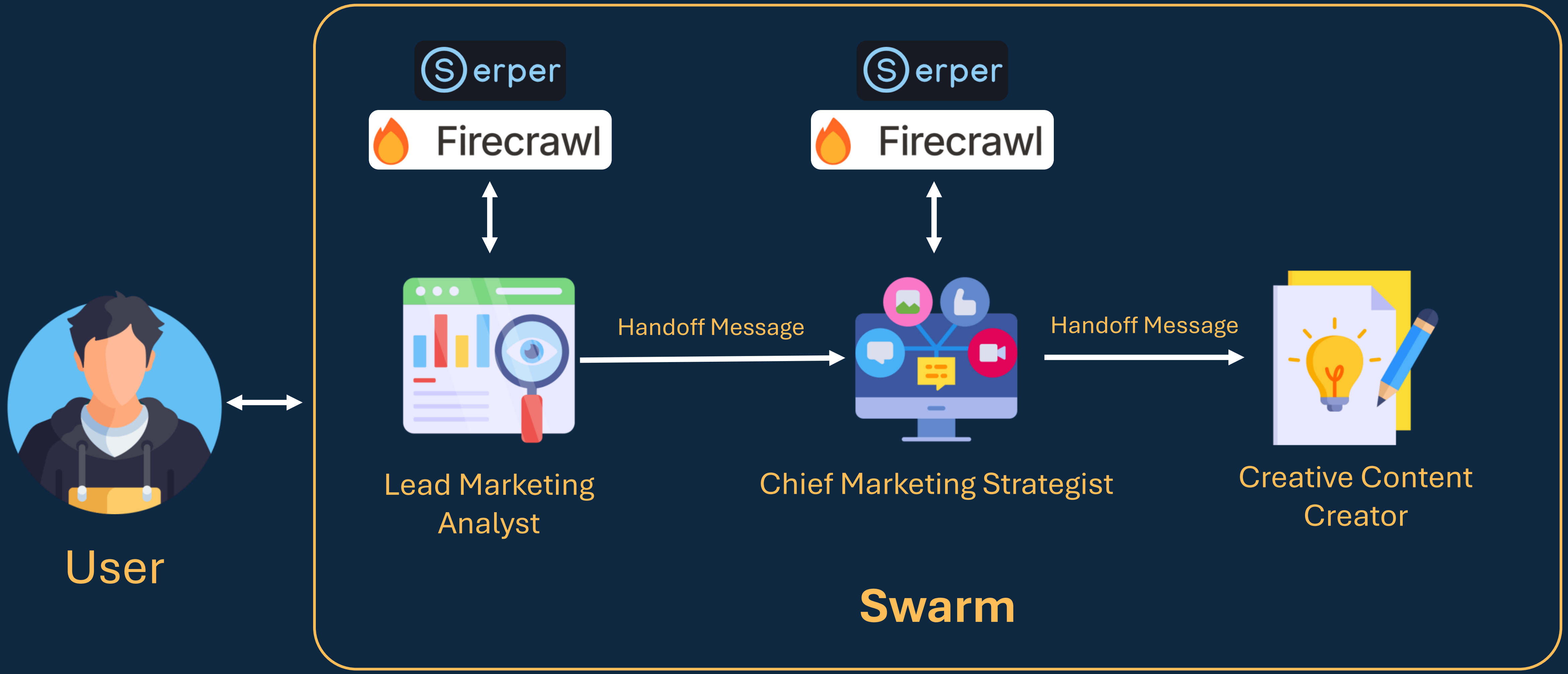
Marketing Campaign Creator – Swarm



Taking AI Agents to Production

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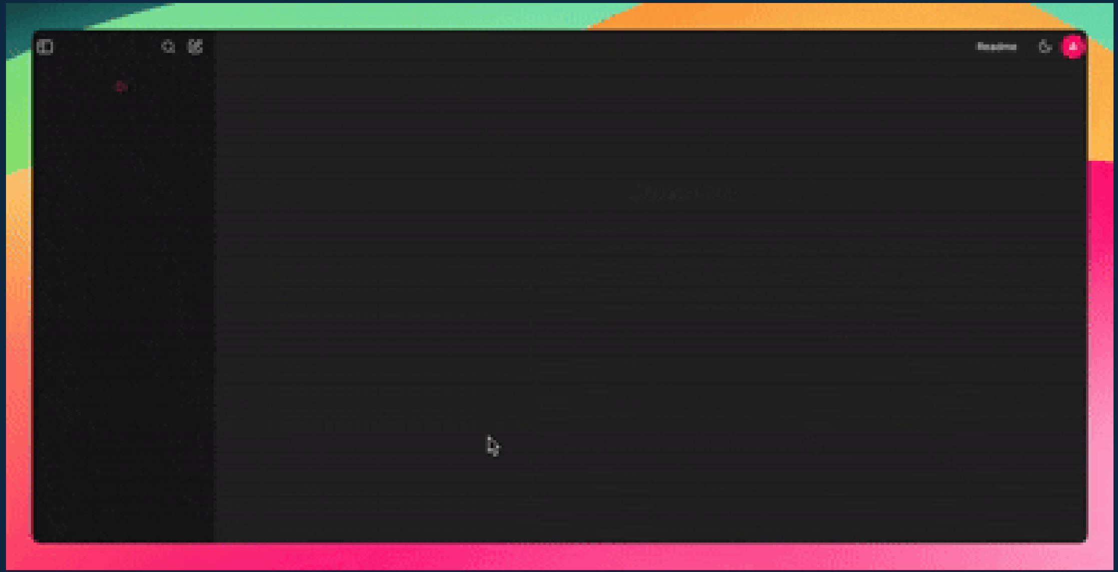
Putting a Face to Agentic Systems – using ChainLit

What is Chainlit?

An open-source Python package to build production-ready Conversational AI.



- [GitHub Repository]
- [Documentation]

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Observability of AI Agents

Transparency

Tracing how inputs transform into outputs & revealing the internal reasoning mechanism

Interpretability

Translating complex AI decisions into human-understandable insights

Continuous Monitoring

Tracking performance, detecting biases & ensuring consistent and reliable AI behavior

Why Observability Matters?

- Enables trust
- Ensures reliability and efficiency
- Facilitates early detection of anomalies

Key Metrics to Track

- Performance Metrics: Response times, latency, and throughput
- Resource Utilization: LLM token usage, cost
- Error Rates: Frequency and types of errors encountered
- User Interactions: Engagement levels and satisfaction scores

Langsmith

AgentOps

OpenTelemetry

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How to Design Resilient AI Agents?

🔄 What if an agent is stuck in a loop?

→ How would you detect and resolve infinite loops or repetitive behavior?

⏳ What if the agent does not respond within the allotted time?

→ How would you handle timeouts and ensure graceful fallback?

📶 What if the agent loses internet connectivity mid-task?

→ Can the agent recover or retry after a network failure?

📥 What if the agent receives incomplete or corrupted input?

→ How would you validate and clean the input before processing?

🔐 What if the agent generates sensitive or harmful content?

→ What safeguards and monitoring mechanisms can you implement?

Time to Build Something Epic...

What Agentic AI System will you build? 🤔

Here are some thoughts to spark off your creative process...

- ⚖️ **AI-Powered Legal Bench**

  Contract Analyst, Legal Researcher, Negotiator, Compliance Agent, Reviewer

- 💹 **Stock Market Hedge Fund**

  Market Analyst, News Sentiment Analyzer, Strategist, Trader, Risk Manager

- 📕 **Book-Writing Crew**

  Plot Architect, Character Developer, Researcher, Writer, Editor


YOU'VE BUILT AGENTS!
NOW LET'S BUILD SMARTER MODELS...